Support Document of *A Robust Neural Control Design for Multi-drone Slung Payload Manipulation with Control Contraction Metrics*

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September 30, 2025

1 Derivation of the system dynamics

This section shows the detailed derivation of our system dynamics using Kane's method. First, define the inertial position of the payload as $\underline{\boldsymbol{x}}_p$, so the multirotor position, velocity, and acceleration in frame I can be defined as following:

$$\begin{aligned} &\underline{\boldsymbol{x}}_j = \underline{\boldsymbol{x}}_p + \underline{\boldsymbol{l}}_j, \quad \underline{\boldsymbol{u}}_j = \dot{\underline{\boldsymbol{x}}}_j = \underline{\boldsymbol{v}}_p + \boldsymbol{B}_j \underline{\boldsymbol{v}}_j, \\ &\underline{\boldsymbol{a}}_j = \dot{\underline{\boldsymbol{u}}} = \dot{\underline{\boldsymbol{v}}}_p + \dot{\boldsymbol{B}}_j \underline{\boldsymbol{v}}_j + \boldsymbol{B}_j \dot{\underline{\boldsymbol{v}}}_j \end{aligned} \tag{1}$$

where $\dot{\boldsymbol{B}}_j$ is the time derivative of $\boldsymbol{B}_j$ and is calculated below:

$$\dot{\boldsymbol{B}}_j = \begin{bmatrix} \boldsymbol{0}_{2\times 2} \\ -\frac{z_j^2 \underline{\boldsymbol{v}}_j^T + \underline{\boldsymbol{r}}_j^T \underline{\boldsymbol{v}}_j \underline{\boldsymbol{r}}_j^T}{z_j^3} \end{bmatrix}. \tag{2}$$

We define the motion variable as $\underline{\boldsymbol{u}} = \begin{bmatrix} \underline{\boldsymbol{v}}_p^T & \underline{\boldsymbol{v}}_1^T & \underline{\boldsymbol{v}}_2^T & \underline{\boldsymbol{v}}_3^T \end{bmatrix}^T$, and the inertial motion variable as $\underline{\boldsymbol{u}}_I = \begin{bmatrix} \underline{\boldsymbol{v}}_p^T & \underline{\boldsymbol{u}}_1^T & \underline{\boldsymbol{u}}_2^T & \underline{\boldsymbol{u}}_3^T \end{bmatrix}^T$, where $\underline{\boldsymbol{v}}_p$ is the payload velocity and $\underline{\boldsymbol{u}}_j$ is the quadrotor velocity, all defined in the inertial frame. We follow Kane's method approach to calculate the partial velocity matrix as below:

$$\begin{aligned} \boldsymbol{U}^T &= \begin{bmatrix} \frac{\partial \underline{\boldsymbol{v}}_p}{\partial \underline{\boldsymbol{v}}_p^T} & \frac{\partial \underline{\boldsymbol{u}}_1}{\partial \underline{\boldsymbol{v}}_p^T} & \frac{\partial \underline{\boldsymbol{u}}_2}{\partial \underline{\boldsymbol{v}}_p^T} & \frac{\partial \underline{\boldsymbol{u}}_3}{\partial \underline{\boldsymbol{v}}_p^T} \\ \frac{\partial \underline{\boldsymbol{v}}_p}{\partial \underline{\boldsymbol{v}}_1^T} & \frac{\partial \underline{\boldsymbol{u}}_1}{\partial \underline{\boldsymbol{v}}_1^T} & \frac{\partial \underline{\boldsymbol{u}}_2}{\partial \underline{\boldsymbol{v}}_1^T} & \frac{\partial \underline{\boldsymbol{u}}_3}{\partial \underline{\boldsymbol{v}}_1^T} \\ \frac{\partial \underline{\boldsymbol{v}}_p}{\partial \underline{\boldsymbol{v}}_2^T} & \frac{\partial \underline{\boldsymbol{u}}_1}{\partial \underline{\boldsymbol{v}}_2^T} & \frac{\partial \underline{\boldsymbol{u}}_2}{\partial \underline{\boldsymbol{v}}_2^T} & \frac{\partial \underline{\boldsymbol{u}}_3}{\partial \underline{\boldsymbol{v}}_2^T} \\ \frac{\partial \underline{\boldsymbol{v}}_p}{\partial \underline{\boldsymbol{v}}_3^T} & \frac{\partial \underline{\boldsymbol{u}}_1}{\partial \underline{\boldsymbol{v}}_3^T} & \frac{\partial \underline{\boldsymbol{u}}_2}{\partial \underline{\boldsymbol{v}}_3^T} & \frac{\partial \underline{\boldsymbol{u}}_3}{\partial \underline{\boldsymbol{v}}_3^T} \end{bmatrix} \\ &= \begin{bmatrix} \boldsymbol{1}_{3\times 3} & \boldsymbol{1}_{3\times 3} & \boldsymbol{1}_{3\times 3} & \boldsymbol{1}_{3\times 3} \\ \boldsymbol{0}_{2\times 3} & \boldsymbol{B}_1^T & \boldsymbol{0}_{2\times 3} & \boldsymbol{0}_{2\times 3} \\ \boldsymbol{0}_{2\times 3} & \boldsymbol{0}_{2\times 3} & \boldsymbol{B}_2^T & \boldsymbol{0}_{2\times 3} \\ \boldsymbol{0}_{2\times 3} & \boldsymbol{0}_{2\times 3} & \boldsymbol{0}_{2\times 3} & \boldsymbol{B}_3^T \end{bmatrix}. \end{aligned} \tag{3}$$

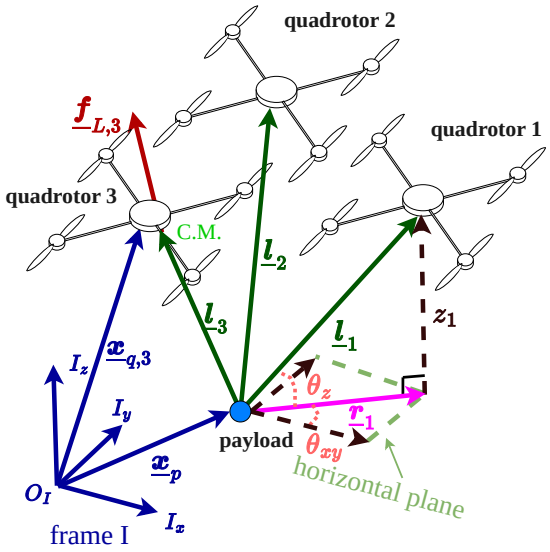


Figure 1: System geometry

Given that the inertial gravity is $\underline{\boldsymbol{g}}_I = \begin{bmatrix} 0 & 0 & -g \end{bmatrix}^T$ with g as the gravitational acceleration, the payload disturbance $\underline{\boldsymbol{\delta}}_p$, the multirotor disturbance $\underline{\boldsymbol{\delta}}_1$, $\underline{\boldsymbol{\delta}}_2$, $\underline{\boldsymbol{\delta}}_3$, and the multirotor three directional lift force $\underline{\boldsymbol{f}}_{L,1}$, $\underline{\boldsymbol{f}}_{L,2}$, $\underline{\boldsymbol{f}}_{L,3}$, the external and internal forces can be formulated:

$$\underline{\boldsymbol{f}} = \overbrace{\begin{bmatrix} m_p \underline{\boldsymbol{g}}_I \\ m_1 \underline{\boldsymbol{g}}_I \\ m_2 \underline{\boldsymbol{g}}_I \\ m_3 \underline{\boldsymbol{g}}_I \end{bmatrix}}^{\underline{\boldsymbol{f}}_g} + \overbrace{\begin{bmatrix} \boldsymbol{0}_{3\times 1} \\ \underline{\boldsymbol{f}}_{L,1} \\ \underline{\boldsymbol{f}}_{L,2} \\ \underline{\boldsymbol{f}}_{L,3} \end{bmatrix}}^{\underline{\boldsymbol{f}}_a} + \overbrace{\begin{bmatrix} \underline{\boldsymbol{\delta}}_p \\ \underline{\boldsymbol{\delta}}_1 \\ \underline{\boldsymbol{\delta}}_2 \\ \underline{\boldsymbol{\delta}}_3 \end{bmatrix}}^{\underline{\boldsymbol{f}}_\delta}, \quad \underline{\boldsymbol{f}}^\star = \begin{bmatrix} -m_p \dot{\underline{\boldsymbol{v}}}_p \\ -m_1 \underline{\boldsymbol{a}}_1 \\ -m_2 \underline{\boldsymbol{a}}_2 \\ -m_3 \underline{\boldsymbol{a}}_3 \end{bmatrix}. \tag{4}$$

We can decompose $\underline{\boldsymbol{f}}^\star$ in order to separate $\dot{\underline{\boldsymbol{u}}}$ and $\underline{\boldsymbol{u}}$, so the Kane's method can be performed accrodingly:

$$\begin{aligned} \underline{\boldsymbol{f}}^\star = & \overbrace{\begin{bmatrix} -m_p \boldsymbol{1}_{3\times 3} & \boldsymbol{0}_{3\times 2} & \boldsymbol{0}_{3\times 2} & \boldsymbol{0}_{3\times 2} \\ -m_1 \boldsymbol{1}_{3\times 3} & -m_1 \boldsymbol{B}_1 & \boldsymbol{0}_{3\times 2} & \boldsymbol{0}_{3\times 2} \\ -m_2 \boldsymbol{1}_{3\times 3} & \boldsymbol{0}_{3\times 2} & -m_2 \boldsymbol{B}_2 & \boldsymbol{0}_{3\times 2} \\ -m_3 \boldsymbol{1}_{3\times 3} & \boldsymbol{0}_{3\times 2} & \boldsymbol{0}_{3\times 2} & -m_3 \boldsymbol{B}_3 \end{bmatrix}}^{\underline{\boldsymbol{f}}_1^\star} \begin{bmatrix} \dot{\underline{\boldsymbol{v}}}_p \\ \dot{\underline{\boldsymbol{v}}}_1 \\ \dot{\underline{\boldsymbol{v}}}_2 \\ \dot{\underline{\boldsymbol{v}}}_3 \end{bmatrix} \\ & + \overbrace{\begin{bmatrix} \boldsymbol{0}_{3\times 3} & \boldsymbol{0}_{3\times 2} & \boldsymbol{0}_{3\times 2} & \boldsymbol{0}_{3\times 2} \\ \boldsymbol{0}_{3\times 3} & -m_1 \dot{\boldsymbol{B}}_1 & \boldsymbol{0}_{3\times 2} & \boldsymbol{0}_{3\times 2} \\ \boldsymbol{0}_{3\times 3} & \boldsymbol{0}_{3\times 2} & -m_2 \dot{\boldsymbol{B}}_2 & \boldsymbol{0}_{3\times 2} \\ \boldsymbol{0}_{3\times 3} & \boldsymbol{0}_{3\times 2} & \boldsymbol{0}_{3\times 2} & -m_3 \dot{\boldsymbol{B}}_3 \end{bmatrix}}^{\underline{\boldsymbol{f}}_2^\star} \begin{bmatrix} \underline{\boldsymbol{v}}_p \\ \underline{\boldsymbol{v}}_1 \\ \underline{\boldsymbol{v}}_2 \\ \underline{\boldsymbol{v}}_3 \end{bmatrix}. \end{aligned} \tag{5}$$

Since Kane's method states that $-\boldsymbol{U}^T \underline{\boldsymbol{f}}^\star = \boldsymbol{M}\underline{\dot{\boldsymbol{u}}} + \boldsymbol{C}\underline{\boldsymbol{u}}$, the inertial matrix $\boldsymbol{M}$ and the gyroscopic matrix $\boldsymbol{C}$, given the total system mass $m_t = m_p + m_1 + m_2 + m_3$, is calculated below:

$$\begin{aligned}\boldsymbol{M} &= -\boldsymbol{U}^T \underline{\boldsymbol{f}}_1^\star \\ &= \begin{bmatrix} m_t \mathbf{1}_{3\times3} & m_1\boldsymbol{B}_1 & m_2\boldsymbol{B}_2 & m_3\boldsymbol{B}_3 \\ m_1\boldsymbol{B}_1^T & m_1\boldsymbol{B}_1^T\boldsymbol{B}_1 & \mathbf{0}_{2\times2} & \mathbf{0}_{2\times2} \\ m_2\boldsymbol{B}_2^T & \mathbf{0}_{2\times2} & m_2\boldsymbol{B}_2^T\boldsymbol{B}_2 & \mathbf{0}_{2\times2} \\ m_3\boldsymbol{B}_3^T & \mathbf{0}_{2\times2} & \mathbf{0}_{2\times2} & m_3\boldsymbol{B}_3^T\boldsymbol{B}_3 \end{bmatrix},\end{aligned} \tag{6}$$

$$\boldsymbol{C} = -\boldsymbol{U}^T \underline{\boldsymbol{f}}_2^\star = \begin{bmatrix} \mathbf{0}_{3\times3} & m_1\dot{\boldsymbol{B}}_1 & m_2\dot{\boldsymbol{B}}_2 & m_3\dot{\boldsymbol{B}}_3 \\ \mathbf{0}_{2\times3} & m_1\boldsymbol{B}_1^T\dot{\boldsymbol{B}}_1 & \mathbf{0}_{2\times2} & \mathbf{0}_{2\times2} \\ \mathbf{0}_{2\times3} & \mathbf{0}_{2\times2} & m_2\boldsymbol{B}_2^T\dot{\boldsymbol{B}}_2 & \mathbf{0}_{2\times2} \\ \mathbf{0}_{2\times3} & \mathbf{0}_{2\times2} & \mathbf{0}_{2\times2} & m_3\boldsymbol{B}_3^T\dot{\boldsymbol{B}}_3 \end{bmatrix}. \tag{7}$$

The external forces of the system, meaning the gravitational force $\underline{\boldsymbol{f}}_g$, the actuation force from the lift of each quadrotor $\underline{\boldsymbol{f}}_a$, and the disturbances $\underline{\boldsymbol{f}}_\delta$, are calculated as below:

$$\begin{aligned}\underline{\boldsymbol{f}}_g &= \boldsymbol{U}^T \underline{\boldsymbol{f}}_{e,1} = \begin{bmatrix} M_t\underline{\boldsymbol{g}}_I \\ m_1\boldsymbol{B}_1^T\underline{\boldsymbol{g}}_I \\ m_2\boldsymbol{B}_2^T\underline{\boldsymbol{g}}_I \\ m_3\boldsymbol{B}_3^T\underline{\boldsymbol{g}}_I \end{bmatrix}, \\ \underline{\boldsymbol{f}}_a &= \boldsymbol{U}^T \underline{\boldsymbol{f}}_{e,2} = \begin{bmatrix} \underline{\boldsymbol{f}}_{L,1} + \underline{\boldsymbol{f}}_{L,2} + \underline{\boldsymbol{f}}_{L,3} \\ \boldsymbol{B}_1^T\underline{\boldsymbol{f}}_{L,1} \\ \boldsymbol{B}_2^T\underline{\boldsymbol{f}}_{L,2} \\ \boldsymbol{B}_3^T\underline{\boldsymbol{f}}_{L,3} \end{bmatrix}, \\ \underline{\boldsymbol{f}}_\delta &= \boldsymbol{U}^T \underline{\boldsymbol{f}}_{e,3} = \begin{bmatrix} \underline{\boldsymbol{\delta}}_p + \underline{\boldsymbol{\delta}}_1 + \underline{\boldsymbol{\delta}}_2 + \underline{\boldsymbol{\delta}}_3 \\ \boldsymbol{B}_1^T\underline{\boldsymbol{\delta}}_1 \\ \boldsymbol{B}_2^T\underline{\boldsymbol{\delta}}_2 \\ \boldsymbol{B}_3^T\underline{\boldsymbol{\delta}}_3 \end{bmatrix}.\end{aligned} \tag{8}$$

After compensating the quadrotor's weights by setting $\underline{\boldsymbol{f}}_{L,j} = -m_j\underline{\boldsymbol{g}}_I + \delta\underline{\boldsymbol{f}}_{L,j}$, we can get $\underline{\boldsymbol{f}}_g + \underline{\boldsymbol{f}}_a = \underline{\boldsymbol{f}}_{g,p} + \boldsymbol{H}\underline{\boldsymbol{\zeta}}$ where

$$\begin{aligned}\underline{\boldsymbol{f}}_{g,p} &= \begin{bmatrix} m_p\underline{\boldsymbol{g}}_I \\ \mathbf{0}_{6\times1} \end{bmatrix} \\ \boldsymbol{H} &= \begin{bmatrix} \mathbf{1}_{3\times3} & \mathbf{1}_{3\times3} & \mathbf{1}_{3\times3} \\ \boldsymbol{B}_1^T & \mathbf{0}_{2\times3} & \mathbf{0}_{2\times3} \\ \mathbf{0}_{2\times3} & \boldsymbol{B}_2^T & \mathbf{0}_{2\times3} \\ \mathbf{0}_{2\times3} & \mathbf{0}_{2\times3} & \boldsymbol{B}_3^T \end{bmatrix} \\ \underline{\boldsymbol{\zeta}} &= \begin{bmatrix} \delta\underline{\boldsymbol{f}}_{L,1} \\ \delta\underline{\boldsymbol{f}}_{L,2} \\ \delta\underline{\boldsymbol{f}}_{L,3} \end{bmatrix}\end{aligned} \tag{9}$$

Also, we can set $\underline{\boldsymbol{f}}_\delta = \boldsymbol{H}_\delta\underline{\boldsymbol{\delta}}$, where

$$\boldsymbol{H}_\delta = \begin{bmatrix} \mathbf{1}_{3\times3} & \mathbf{1}_{3\times3} & \mathbf{1}_{3\times3} & \mathbf{1}_{3\times3} \\ \mathbf{0}_{2\times3} & \boldsymbol{B}_1^T & \mathbf{0}_{2\times3} & \mathbf{0}_{2\times3} \\ \mathbf{0}_{2\times3} & \mathbf{0}_{2\times3} & \boldsymbol{B}_2^T & \mathbf{0}_{2\times3} \\ \mathbf{0}_{2\times3} & \mathbf{0}_{2\times3} & \mathbf{0}_{2\times3} & \boldsymbol{B}_3^T \end{bmatrix}$$
$$\underline{\boldsymbol{\delta}} = \begin{bmatrix} \underline{\boldsymbol{\delta}}_p \\ \underline{\boldsymbol{\delta}}_1 \\ \underline{\boldsymbol{\delta}}_2 \\ \underline{\boldsymbol{\delta}}_3 \end{bmatrix}. \tag{10}$$

As Kane's method states that

$$\boldsymbol{U}^T(\underline{\boldsymbol{f}} + \underline{\boldsymbol{f}}^*) = \mathbf{0}, \tag{11}$$

we get the following equation of motion

$$\boldsymbol{M}\dot{\underline{\boldsymbol{u}}} + \boldsymbol{C}\underline{\boldsymbol{u}} = \underline{\boldsymbol{f}}_{g,p} + \boldsymbol{H}\underline{\boldsymbol{\zeta}} + \boldsymbol{H}_\delta\underline{\boldsymbol{\delta}} \tag{12}$$

2 Derivation of the UDE update law

If we analyze the kth row block of the equations of motion in (12) ($k = 2,3,4$), we have the expression of **(18) in our paper**. Take $k = 2$ as an example:

$$\begin{aligned} &\begin{bmatrix} m_1\boldsymbol{B}_1^T & m_1\boldsymbol{B}_1^T\boldsymbol{B}_1 & \mathbf{0}_{2\times2} & \mathbf{0}_{2\times2} \end{bmatrix}\dot{\underline{\boldsymbol{u}}} + \begin{bmatrix} \mathbf{0}_{2\times3} & m_1\boldsymbol{B}_1^T\dot{\boldsymbol{B}}_1 & \mathbf{0}_{2\times2} & \mathbf{0}_{2\times2} \end{bmatrix}\underline{\boldsymbol{u}} \\ &= \boldsymbol{B}_1^T\delta\underline{\boldsymbol{f}}_{L,1} + \boldsymbol{B}_1^T\underline{\boldsymbol{\delta}}_1 \end{aligned} \tag{13}$$

Applying the expression of $\dot{\underline{\boldsymbol{u}}}$ and $\underline{\boldsymbol{u}}$, we have the following:

$$\begin{aligned} &m_1\boldsymbol{B}_1^T(\dot{\underline{\boldsymbol{v}}}_p + \boldsymbol{B}_1\dot{\underline{\boldsymbol{v}}}_1) + m_1\boldsymbol{B}_1^T\dot{\boldsymbol{B}}_1\underline{\boldsymbol{v}}_1 = m_1\boldsymbol{B}_1^T(\dot{\underline{\boldsymbol{v}}}_p + \boldsymbol{B}_1\dot{\underline{\boldsymbol{v}}}_1 + \dot{\boldsymbol{B}}_1\underline{\boldsymbol{v}}_1) \\ &= \boldsymbol{B}_1^T\delta\underline{\boldsymbol{f}}_{L,1} + \boldsymbol{B}_1^T\underline{\boldsymbol{\delta}}_1. \end{aligned} \tag{14}$$

Note that the inertial velocity of the jth quadrotor is $\underline{\boldsymbol{v}}_{q,j}$:

$$\underline{\boldsymbol{v}}_{q,1} = \underline{\boldsymbol{v}}_p + \boldsymbol{B}_1\underline{\boldsymbol{v}}_1 \tag{15}$$

Hence we have the following:

$$m_1\boldsymbol{B}_1^T\frac{d\underline{\boldsymbol{v}}_{q,1}}{dt} = \boldsymbol{B}_1^T(\delta\underline{\boldsymbol{f}}_{L,1} + \underline{\boldsymbol{\delta}}_1). \tag{16}$$

It is easy to repeat the above with $k = 3,4$ to get **(18) in our paper**.

If we analyze the first row block of the equations of motion in (12), we have the following:

$$\begin{aligned} &\begin{bmatrix} m_t\mathbf{1}_{3\times3} & m_1\boldsymbol{B}_1 & m_2\boldsymbol{B}_2 & m_3\boldsymbol{B}_3 \end{bmatrix}\dot{\underline{\boldsymbol{u}}} + \begin{bmatrix} \mathbf{0}_{3\times3} & m_1\dot{\boldsymbol{B}}_1 & m_2\dot{\boldsymbol{B}}_2 & m_3\dot{\boldsymbol{B}}_3 \end{bmatrix}\underline{\boldsymbol{u}} \\ &= m_p\underline{\boldsymbol{g}}_I + \sum_{j=1}^{N}\delta\underline{\boldsymbol{f}}_{L,j} + \underline{\boldsymbol{\delta}}_p + \sum_{j=1}^{N}\underline{\boldsymbol{\delta}}_j \end{aligned} \tag{17}$$

where $N = 3$. Then we apply the error dynamics in **(23)** and the definition of $\underline{\boldsymbol{\delta}}_T$ in **(16) of our paper**, we get the following:

$$
\begin{aligned}
&d(m_t \underline{\boldsymbol{v}}_p + \sum_{j=1}^{N} m_j \boldsymbol{B}_j \underline{\boldsymbol{v}}_j)/dt = \hat{\underline{\boldsymbol{\delta}}}_T - \tilde{\underline{\boldsymbol{\delta}}}_T + m_p \underline{\boldsymbol{g}}_I + \sum_{j=1}^{N} (\underline{\boldsymbol{f}}_{L,j} + \underline{\boldsymbol{\delta}}_{\perp,j}) \\
&= \hat{\underline{\boldsymbol{\delta}}}_T + \dot{\tilde{\underline{\boldsymbol{\delta}}}}_T/\lambda_T + \sum_{j=1}^{N} \tilde{\underline{\boldsymbol{\delta}}}_{\perp,j} + m_p \underline{\boldsymbol{g}}_I + \sum_{j=1}^{N} (\underline{\boldsymbol{f}}_{L,j} + \underline{\boldsymbol{\delta}}_{\perp,j}) \\
&= \hat{\underline{\boldsymbol{\delta}}}_T + \dot{\tilde{\underline{\boldsymbol{\delta}}}}_T/\lambda_T + m_p \underline{\boldsymbol{g}}_I + \sum_{j=1}^{N} (\underline{\boldsymbol{f}}_{L,j} + \underline{\boldsymbol{\delta}}_{\perp,j} + \tilde{\underline{\boldsymbol{\delta}}}_{\perp,j}) \\
&= \hat{\underline{\boldsymbol{\delta}}}_T + \dot{\hat{\underline{\boldsymbol{\delta}}}}_T/\lambda_T + m_p \underline{\boldsymbol{g}}_I + \sum_{j=1}^{N} (\underline{\boldsymbol{f}}_{L,j} + \hat{\underline{\boldsymbol{\delta}}}_{\perp,j}) \\
&\Rightarrow \dot{\hat{\underline{\boldsymbol{\delta}}}}_T/\lambda_T = d(m_t \underline{\boldsymbol{v}}_p + \sum_{j=1}^{N} m_j \boldsymbol{B}_j \underline{\boldsymbol{v}}_j)/dt - \hat{\underline{\boldsymbol{\delta}}}_T - m_p \underline{\boldsymbol{g}}_I - \sum_{j=1}^{N} (\underline{\boldsymbol{f}}_{L,j} + \hat{\underline{\boldsymbol{\delta}}}_{\perp,j})
\end{aligned}
\tag{18}
$$

3 Implementation of the attitude tracking controller

After obtaining $\underline{\boldsymbol{\zeta}}_c$ from the UDE, the total lift force from each drone $\underline{\boldsymbol{f}}_{Lc,j}$ is calculated as the following:

$$
\underline{\boldsymbol{f}}_{Lc,j} = \underline{\boldsymbol{\zeta}}_{c,j} - m_j \underline{\boldsymbol{g}}_I, \tag{19}
$$

where $\underline{\boldsymbol{\zeta}}_c = \begin{bmatrix} \underline{\boldsymbol{\zeta}}_{c,1}^T & \underline{\boldsymbol{\zeta}}_{c,2}^T & \underline{\boldsymbol{\zeta}}_{c,3}^T \cdot \end{bmatrix}^T$ The desired rotation $\underline{\boldsymbol{R}}_{Ij,d}$ to achieve $\underline{\boldsymbol{f}}_{Lc,j}$ is then calculated in **Equation (29) of our paper**. $\underline{\boldsymbol{\omega}}_d$ and $\dot{\underline{\boldsymbol{\omega}}}_d$ is then obtained by numerical differentiation of $\underline{\boldsymbol{R}}_{Ij,d}$. Under the attitude control law stated in **Equation (30) of our paper**, we can obtain the attitude dynamics of each quadrotor $\underline{\boldsymbol{R}}_{Ij}$ and $\underline{\boldsymbol{\omega}}_j$. Lastly, we can obtain the lift force on each drone as the following:

$$
\underline{\boldsymbol{f}}_{L,j} = \underline{\boldsymbol{R}}_{Ij} \begin{bmatrix} 0 \\ 0 \\ ||\underline{\boldsymbol{f}}_{Lc,j}|| \end{bmatrix} = \underline{\boldsymbol{R}}_{Ij} \underline{\boldsymbol{e}}_3 f_j. \tag{20}
$$

Hence, our feedback control signal is written as

$$
\underline{\boldsymbol{\zeta}} = vstack(\underline{\boldsymbol{f}}_{L,j} + m_j \underline{\boldsymbol{g}}_I) = vstack(\underline{\boldsymbol{R}}_{Ij} \underline{\boldsymbol{e}}_3 f_j + m_j \underline{\boldsymbol{g}}_I) \tag{21}
$$

4 Stability Analysis

4.1 Disturbance on the closed-loop system

Equation (36) of our paper states the following:

$$
\underline{\boldsymbol{d}} = \boldsymbol{G}(\underline{\boldsymbol{x}}) \underline{\boldsymbol{\zeta}}_e + \boldsymbol{G}_\delta(\underline{\boldsymbol{x}}) \underline{\boldsymbol{\delta}}_e \tag{22}
$$

where

$$\boldsymbol{G}(\underline{\boldsymbol{x}}) = \begin{bmatrix} \mathbf{0}_{9\times 9} \\ \boldsymbol{M}^{-1}\boldsymbol{H} \end{bmatrix}, \quad \boldsymbol{G}_{\delta}(\underline{\boldsymbol{x}}) = \begin{bmatrix} \mathbf{0}_{9\times 12} \\ \boldsymbol{M}^{-1}\boldsymbol{H}_{\delta} \end{bmatrix}. \tag{23}$$

$\underline{\boldsymbol{\zeta}}_e$ is the error between the target force generated by our control law and the actual force provided by the attitude tracker in Section V of our paper. Here, we provide a detailed derivation of the disturbance caused by the UDE estimation error: $\boldsymbol{G}_{\delta}(\underline{\boldsymbol{x}})\underline{\boldsymbol{\delta}}_e$. According to (9) and **(28) in our paper**, we get the following:

$$\boldsymbol{H}\underline{\boldsymbol{f}}_{\delta} = \begin{bmatrix} \mathbf{1}_{3\times 3} & \mathbf{1}_{3\times 3} & \mathbf{1}_{3\times 3} \\ \boldsymbol{B}_1^T & \mathbf{0}_{2\times 3} & \mathbf{0}_{2\times 3} \\ \mathbf{0}_{2\times 3} & \boldsymbol{B}_2^T & \mathbf{0}_{2\times 3} \\ \mathbf{0}_{2\times 3} & \mathbf{0}_{2\times 3} & \boldsymbol{B}_3^T \end{bmatrix} \begin{bmatrix} -\underline{\boldsymbol{n}}_1\hat{\delta}_{T,1} - \hat{\underline{\boldsymbol{\delta}}}_{\perp,1} \\ -\underline{\boldsymbol{n}}_2\hat{\delta}_{T,2} - \hat{\underline{\boldsymbol{\delta}}}_{\perp,2} \\ -\underline{\boldsymbol{n}}_3\hat{\delta}_{T,3} - \hat{\underline{\boldsymbol{\delta}}}_{\perp,3} \end{bmatrix} \tag{24}$$

From (28) in our paper, we know that:

$$\begin{bmatrix} \hat{\delta}_{T,1} & \hat{\delta}_{T,2} & \hat{\delta}_{T,3} \end{bmatrix}^T = \begin{bmatrix} \underline{\boldsymbol{n}}_1, \underline{\boldsymbol{n}}_2, \underline{\boldsymbol{n}}_3 \end{bmatrix}^{-1} \hat{\underline{\boldsymbol{\delta}}}_T \rightarrow \begin{bmatrix} \underline{\boldsymbol{n}}_1, \underline{\boldsymbol{n}}_2, \underline{\boldsymbol{n}}_3 \end{bmatrix} \begin{bmatrix} \hat{\delta}_{T,1} \\ \hat{\delta}_{T,2} \\ \hat{\delta}_{T,3} \end{bmatrix} = \hat{\underline{\boldsymbol{\delta}}}_T \tag{25}$$

In addition, $\boldsymbol{B}_j^T\underline{\boldsymbol{n}}_j = \underline{\mathbf{0}}$. Hence we have the following:

$$\boldsymbol{H}\underline{\boldsymbol{f}}_{\delta} = \begin{bmatrix} -\hat{\underline{\boldsymbol{\delta}}}_T - \sum_{j=1}^{3}\hat{\underline{\boldsymbol{\delta}}}_{\perp,j} \\ -\boldsymbol{B}_1^T\hat{\underline{\boldsymbol{\delta}}}_{\perp,1} \\ -\boldsymbol{B}_2^T\hat{\underline{\boldsymbol{\delta}}}_{\perp,2} \\ -\boldsymbol{B}_3^T\hat{\underline{\boldsymbol{\delta}}}_{\perp,3} \end{bmatrix} \tag{26}$$

Therefore, the effect of the disturbance estimation error has the following form:

$$\boldsymbol{H}_{\delta}\underline{\boldsymbol{\delta}}_e = \boldsymbol{H}_{\delta}\underline{\boldsymbol{\delta}} + \boldsymbol{H}\underline{\boldsymbol{f}}_{\delta} = \begin{bmatrix} -\hat{\underline{\boldsymbol{\delta}}}_T - \sum_{j=1}^{3}\hat{\underline{\boldsymbol{\delta}}}_{\perp,j} \\ -\boldsymbol{B}_1^T\hat{\underline{\boldsymbol{\delta}}}_{\perp,1} \\ -\boldsymbol{B}_2^T\hat{\underline{\boldsymbol{\delta}}}_{\perp,2} \\ -\boldsymbol{B}_3^T\hat{\underline{\boldsymbol{\delta}}}_{\perp,3} \end{bmatrix} + \begin{bmatrix} \underline{\boldsymbol{\delta}}_p + \sum_{j=1}^{3}\underline{\boldsymbol{\delta}}_j \\ \boldsymbol{B}_1^T\underline{\boldsymbol{\delta}}_1 \\ \boldsymbol{B}_2^T\underline{\boldsymbol{\delta}}_2 \\ \boldsymbol{B}_3^T\underline{\boldsymbol{\delta}}_3 \end{bmatrix}. \tag{27}$$

Since $\underline{\boldsymbol{\delta}}_p + \sum_{j=1}^{N}\underline{\boldsymbol{\delta}}_j = \underline{\boldsymbol{\delta}}_T + \sum_{j=1}^{N}\underline{\boldsymbol{\delta}}_{\perp,j}$ and $\boldsymbol{B}_j^T\underline{\boldsymbol{\delta}}_{\parallel,j} = \mathbf{0}$, we have the following:

$$\begin{aligned} \boldsymbol{H}_{\delta}\underline{\boldsymbol{\delta}}_e = \boldsymbol{H}_{\delta}\underline{\boldsymbol{\delta}} + \boldsymbol{H}\underline{\boldsymbol{f}}_{\delta} &= \begin{bmatrix} -\hat{\underline{\boldsymbol{\delta}}}_T - \sum_{j=1}^{3}\hat{\underline{\boldsymbol{\delta}}}_{\perp,j} \\ -\boldsymbol{B}_1^T\hat{\underline{\boldsymbol{\delta}}}_{\perp,1} \\ -\boldsymbol{B}_2^T\hat{\underline{\boldsymbol{\delta}}}_{\perp,2} \\ -\boldsymbol{B}_3^T\hat{\underline{\boldsymbol{\delta}}}_{\perp,3} \end{bmatrix} + \begin{bmatrix} \underline{\boldsymbol{\delta}}_T + \sum_{j=1}^{3}\underline{\boldsymbol{\delta}}_{\perp,j} \\ \boldsymbol{B}_1^T\underline{\boldsymbol{\delta}}_{\perp,1} \\ \boldsymbol{B}_2^T\underline{\boldsymbol{\delta}}_{\perp,2} \\ \boldsymbol{B}_3^T\underline{\boldsymbol{\delta}}_{\perp,3} \end{bmatrix} \\ = \begin{bmatrix} -\tilde{\underline{\boldsymbol{\delta}}}_T - \sum_{j=1}^{3}\tilde{\underline{\boldsymbol{\delta}}}_{\perp,j} \\ -\boldsymbol{B}_1^T\tilde{\underline{\boldsymbol{\delta}}}_{\perp,1} \\ -\boldsymbol{B}_2^T\tilde{\underline{\boldsymbol{\delta}}}_{\perp,2} \\ -\boldsymbol{B}_3^T\tilde{\underline{\boldsymbol{\delta}}}_{\perp,3} \end{bmatrix} &= \boldsymbol{H}_{\delta}\begin{bmatrix} -\tilde{\underline{\boldsymbol{\delta}}}_T \\ -\tilde{\underline{\boldsymbol{\delta}}}_{\perp,1} \\ -\tilde{\underline{\boldsymbol{\delta}}}_{\perp,2} \\ -\tilde{\underline{\boldsymbol{\delta}}}_{\perp,3} \end{bmatrix} = \boldsymbol{H}_{\delta}\underline{\boldsymbol{\delta}}_e \end{aligned} \tag{28}$$

where $\underline{\boldsymbol{\delta}}_e = -[\underline{\tilde{\boldsymbol{\delta}}}_T^T, \underline{\tilde{\boldsymbol{\delta}}}_{\perp,1}^T, \underline{\tilde{\boldsymbol{\delta}}}_{\perp,2}^T, \underline{\tilde{\boldsymbol{\delta}}}_{\perp,3}^T]^T$. Hence, the disturbance to the closed-loop system caused by the estimation error $\boldsymbol{G}_\delta(\underline{\boldsymbol{x}})\underline{\boldsymbol{\delta}}_e$ in **(36) of our paper** is:

$$\begin{aligned} &\boldsymbol{G}(\underline{\boldsymbol{x}})\underline{\boldsymbol{f}}_\delta + \boldsymbol{G}_\delta(\underline{\boldsymbol{x}})\underline{\boldsymbol{\delta}} \\ &= \begin{bmatrix} \boldsymbol{0}_{9\times 1} \\ \boldsymbol{M}^{-1}\boldsymbol{H}\underline{\boldsymbol{f}}_\delta \end{bmatrix} + \begin{bmatrix} \boldsymbol{0}_{9\times 1} \\ \boldsymbol{M}^{-1}\boldsymbol{H}_\delta\underline{\boldsymbol{\delta}} \end{bmatrix} = \begin{bmatrix} \boldsymbol{0}_{9\times 1} \\ \boldsymbol{M}^{-1}\boldsymbol{H}_\delta\underline{\boldsymbol{\delta}}_e \end{bmatrix} = \boldsymbol{G}_\delta(\underline{\boldsymbol{x}})\underline{\boldsymbol{\delta}}_e \end{aligned} \tag{29}$$

4.2 The Uniformly Continuous Property of $\dot{V}_e$

According to **(34) of our paper**, $\dot{V}_e$ is:

$$\dot{V}_e = -\sum_{j=1}^{N} \underline{\tilde{\boldsymbol{z}}}_j^T \begin{bmatrix} c_T\lambda_T/N\cdot\boldsymbol{1} & c_T\lambda_T/2\cdot\boldsymbol{1} \\ c_T\lambda_T/2\cdot\boldsymbol{1} & c_T\lambda_T N/2 + c_j\kappa_j/N\cdot\boldsymbol{1} \end{bmatrix} \underline{\tilde{\boldsymbol{z}}}_j = -\sum_{j=1}^{N} \underline{\tilde{\boldsymbol{z}}}_j^T \boldsymbol{\Sigma} \underline{\tilde{\boldsymbol{z}}}_j \tag{30}$$

Note that $\boldsymbol{\Sigma}$ is constant. Then $\ddot{V}_e$ is

$$\ddot{V}_e = -2\sum_{j=1}^{N} \underline{\tilde{\boldsymbol{z}}}_j^T \boldsymbol{\Sigma} \underline{\dot{\tilde{\boldsymbol{z}}}}_j \tag{31}$$

$$\begin{aligned} \underline{\dot{\tilde{\boldsymbol{z}}}}_j &= \begin{bmatrix} \underline{\dot{\tilde{\boldsymbol{\delta}}}}_T \\ \underline{\dot{\tilde{\boldsymbol{\delta}}}}_{\perp,j} \end{bmatrix} = \begin{bmatrix} \lambda_T(-\underline{\tilde{\boldsymbol{\delta}}}_T - \sum_{j=1}^{N}\underline{\tilde{\boldsymbol{\delta}}}_{\perp,j}) \\ (\boldsymbol{1} - \underline{\boldsymbol{l}}_j\underline{\boldsymbol{l}}_j^T/l^2)\underline{\dot{\tilde{\boldsymbol{\delta}}}}_j + (-\boldsymbol{B}_j\underline{\boldsymbol{v}}_j\underline{\boldsymbol{l}}_j^T/l^2 - \underline{\boldsymbol{l}}_j\underline{\boldsymbol{v}}_j^T\boldsymbol{B}_j^T/l^2)\underline{\tilde{\boldsymbol{\delta}}}_j \end{bmatrix} \\ &= \begin{bmatrix} \lambda_T(-\underline{\tilde{\boldsymbol{\delta}}}_T - \sum_{j=1}^{N}\underline{\tilde{\boldsymbol{\delta}}}_{\perp,j}) \\ -(\boldsymbol{1} - \underline{\boldsymbol{l}}_j\underline{\boldsymbol{l}}_j^T/l^2)\kappa_j\mathfrak{B}_j\underline{\tilde{\boldsymbol{\delta}}}_{\perp,j} + (-\boldsymbol{B}_j\underline{\boldsymbol{v}}_j\underline{\boldsymbol{l}}_j^T/l^2 - \underline{\boldsymbol{l}}_j\underline{\boldsymbol{v}}_j^T\boldsymbol{B}_j^T/l^2)\underline{\tilde{\boldsymbol{\delta}}}_j \end{bmatrix} \end{aligned} \tag{32}$$

Since $\underline{\tilde{\boldsymbol{\delta}}}_T$, $\underline{\tilde{\boldsymbol{\delta}}}_{\perp,j}$, $\underline{\tilde{\boldsymbol{\delta}}}_j$, and $\underline{\boldsymbol{v}}_j$ are bounded, $\underline{\dot{\tilde{\boldsymbol{z}}}}_j$ is bounded. Hence, $\ddot{V}_e$ is bounded, and $\dot{V}_e$ is uniformly continuous.

5 Additional Simulation Results

The performance of the circular and figure-8 path following is demonstrated in Fig.3 and Fig.4. The disturbance force is a summation of constant noise and stochastic noise $\underline{\boldsymbol{\delta}} = \underline{\boldsymbol{\delta}}_c + \underline{\boldsymbol{\delta}}_s$, where $\underline{\boldsymbol{\delta}}_c = \begin{bmatrix} 0.3 & -0.2 & 0.5 & 0.3 & \dots & 0.3 \end{bmatrix}^T \in \mathbb{R}^{12}$. Only a constant noise is used in circular path following to ensure perfect tracking when the noise level is constant. A stochastic noise of $\underline{\boldsymbol{\delta}}_s \sim \mathscr{U}(0,1)\cdot 0.3$ is used in figure-8 path following. Accuracy is significantly improved in both cases with the UDE turned on, and the system can quickly converge to a bounded neighbourhood of the reference trajectory for the figure-8 configuration. For the circular path, the noise converges to 0, satisfying **Assumption 1 in our paper**.

References

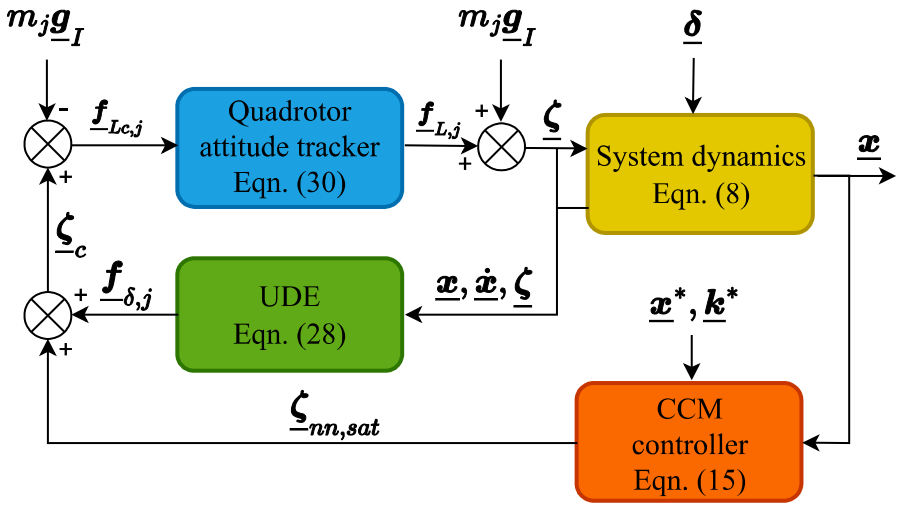
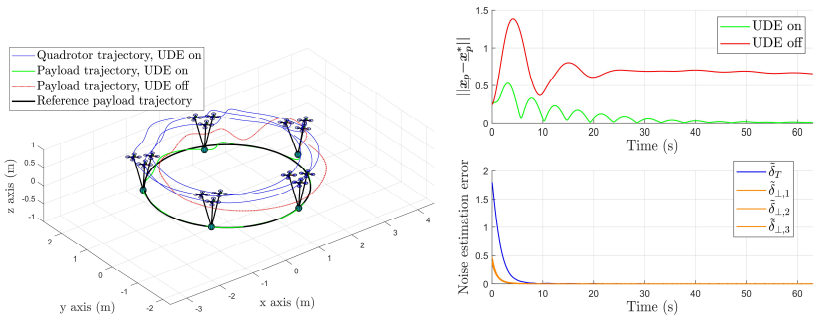


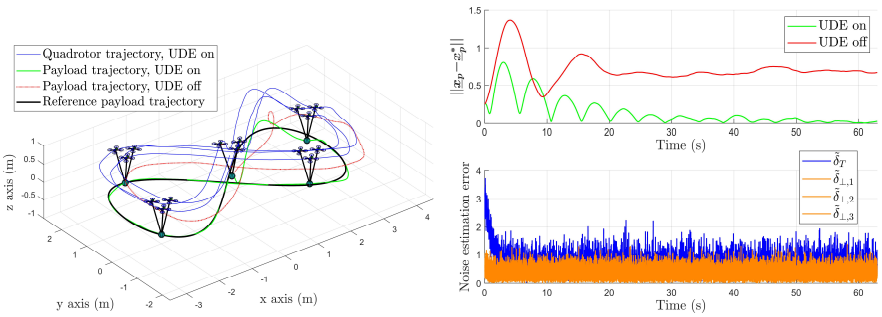
Figure 2: Block diagram of the complete system.



(a) Circular trajectory

(b) Circular path UDE error

Figure 3: Circular path MATLAB simulation performance plots, (a): comparison with UDE on and off; (b): estimated disturbance error with respect to time when using UDE.



(a) Figure-8 trajectory

(b) Figure-8 path UDE error

Figure 4: Figure-8 path MATLAB simulation performance plots, (a): comparison with UDE on and off; (b): estimated disturbance error with respect to time when using UDE.